



How physicians answer “fair assessment” survey questions

Analysis of fairness ratings, survey comments & interviews

Disclosure

- Nothing to disclose

Team members

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Overview of project

Since 2007, ABIM has routinely asked physicians to rate whether assessments are “a fair assessment of clinical knowledge in this discipline.” What do their responses mean?

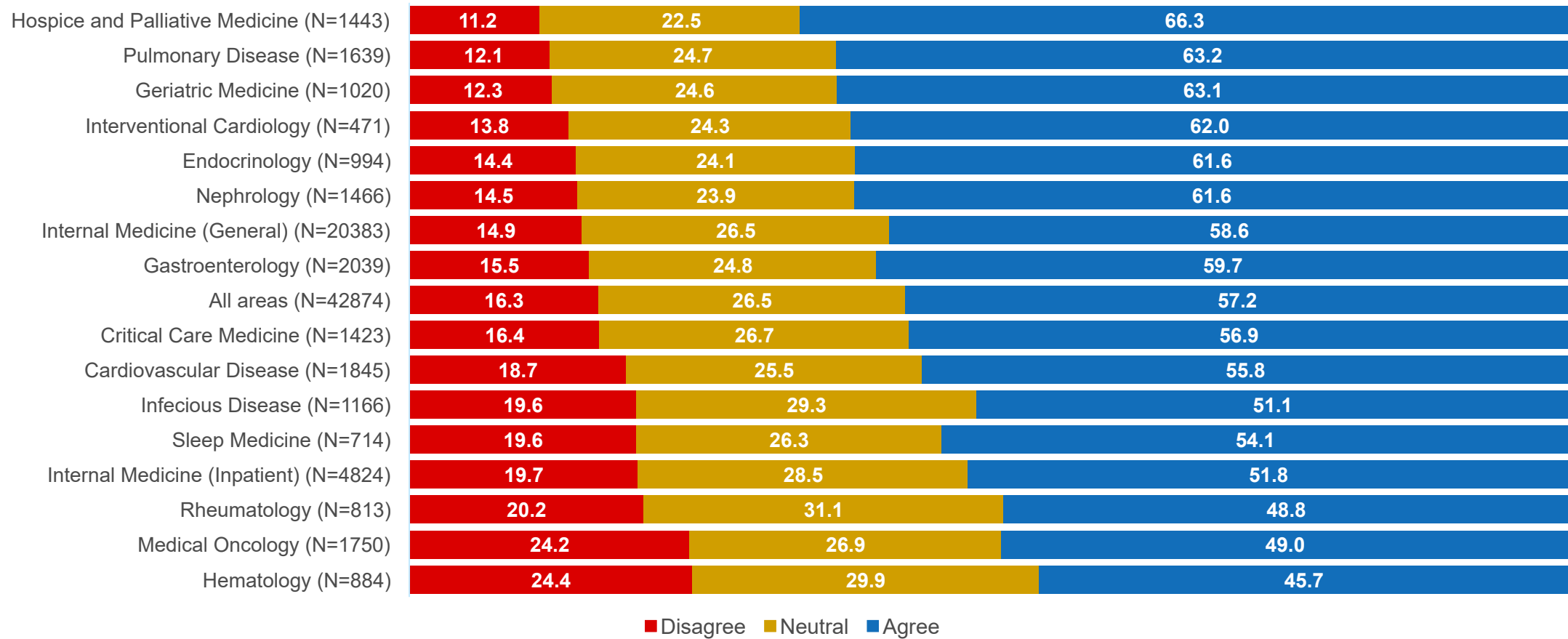
Intended as gestalt of 3 domains related specifically to assessment content

- Content of individual questions – too many questions that felt irrelevant/esoteric?
- Scope of content – right topics included to represent knowledge in the specialty?
- Difficulty – appropriately difficult?

We examined:

- **Comments** data from recent Longitudinal Knowledge Assessment (LKA) and long-form Maintenance of Certification (MOC) exam surveys
- **Interviews** with recent survey respondents (Internal Medicine and Medical Oncology LKA takers)
- Correlation between **LKA performance** and fairness ratings

“The LKA is a fair assessment of clinical knowledge in this discipline” survey responses*



*Based on the 2024 LKA End-of-Year Evaluation Survey, which had a 61% response rate. Numbers may not add up to exactly 100% due to rounding.

Analysis of sample survey comments

Analysis of sample comments data from recent LKA and long-form MOC exam surveys

- LKA: random sample of 500 comments from 2023 LKA end-of-year survey (300 disagree, 200 neither agree/disagree)
- Full-day MOC exam: random sample of about 300 comments from MOC exams 2019-24

3 major themes of LKA & longform survey comments –

- **Specific questions** - wrong “correct” answer; missing necessary info; written to be inappropriately “tricky”
- **Scope of content** - too specialized / from other specialties; esoteric; missing “bread & butter”
- **Format/requirement** - limited time; MCQs; summative (not just for learning); required (cert. at stake)

Respondents w/ different ratings can provide very similar comments → “Similar criteria, different standards”?

Physician interviews

Interviews with 61 2024 end-of-year survey respondents from 2 different LKAs

- 43 Internal Medicine-only LKA takers, 18 Medical Oncology LKA takers
- 3 “fair assessment” response groups: “agree” / “neutral” / “disagree”

Chosen for several reasons – generalists / specialists; **range of sentiments** on fairness

- Internal Medicine: 58.6% agree / 26.5% neutral / **14.9% disagree**
- Medical Oncology: 49.0% agree / 26.9% neutral / **24.2% disagree**

Interview topics:

- How did they arrive at their “fairness” response?
- What could have made LKA more fair?
- Fairness of LKA vs other assessments?
- What makes an assessment “fair”, in general?

Physician interviews – findings

Dimensions of fairness cited (similar to survey comments):

- proportion of problematic questions; scope of content; fairness of LKA format; MOC requirement

Some **specialty-specific** dimensions

- IM: **inpatient / outpatient** content mix
- Medical Oncology: whether answers were **out of date** with latest evidence

Like survey respondents, **interviewees could have markedly different experiences of same assessment**

- **“Wrong” questions** (some thought these were acceptable, in proportion)
- **Limited time per question** (some thought this felt very realistic to practice)
- Being asked questions **across scope of discipline**, or about **rare conditions**:
 - “For specialty questions, it’s a good check for your referral basis skills”* (IM positive)
 - “80% of questions should be what we see in normal everyday practice and 20% can be ‘weird’ questions.”* (IM neutral)

Physician interviews – findings

Interviewees may have **changed their fairness rating** since the last survey

- (usually, from negative to neutral/positive, citing greater familiarity w/format, online/resources setup, etc.)

A common complaint - **“tricky” / “gotcha” / “guess what I’m thinking” questions**

- Ambiguous - are these on “important but rare” topics that physicians happen to get wrong?

“Some questions seem there just to push scores down, or into a bell curve... ‘gotcha’ questions with hidden details in the wording”... (but also “it’s true that some of these details you shouldn’t miss in practice”) (Med Onc neutral)

Expectations for MOC assessment vs initial certification – **physicians can question the value of what is “in” the assessment, given their awareness of what is “out”**

- Practice specialization, local context, outcomes, competencies other than knowledge (teamwork, communication, etc.)

Statistical analysis – responses / performance

We looked at how IM and Med Onc LKA takers' fairness responses relate to their LKA performance:

- End of 2024 LKA Survey – fairness question responses
- End of 2024 LKA standardized scores
 - With or without Progress Reports – 2022-3 LKA cohorts vs 2024 cohort

RESULTS

- There is a weak positive relationship between LKA standardized scores and fairness ratings
- **Does not** differ meaningfully between LKA takers who received progress reports and those who did not
- Factors other than participants' **LKA performance** may play a more substantial role in shaping their views on fairness

“Fair assessment” responders’ gestalt

Observations:

- Individuals combine / balance factors differently
- Informed by attitudes toward ABIM & MOC
- There can be discipline-specific concerns
- Performance does not seem to unduly drive fairness ratings



Conclusions

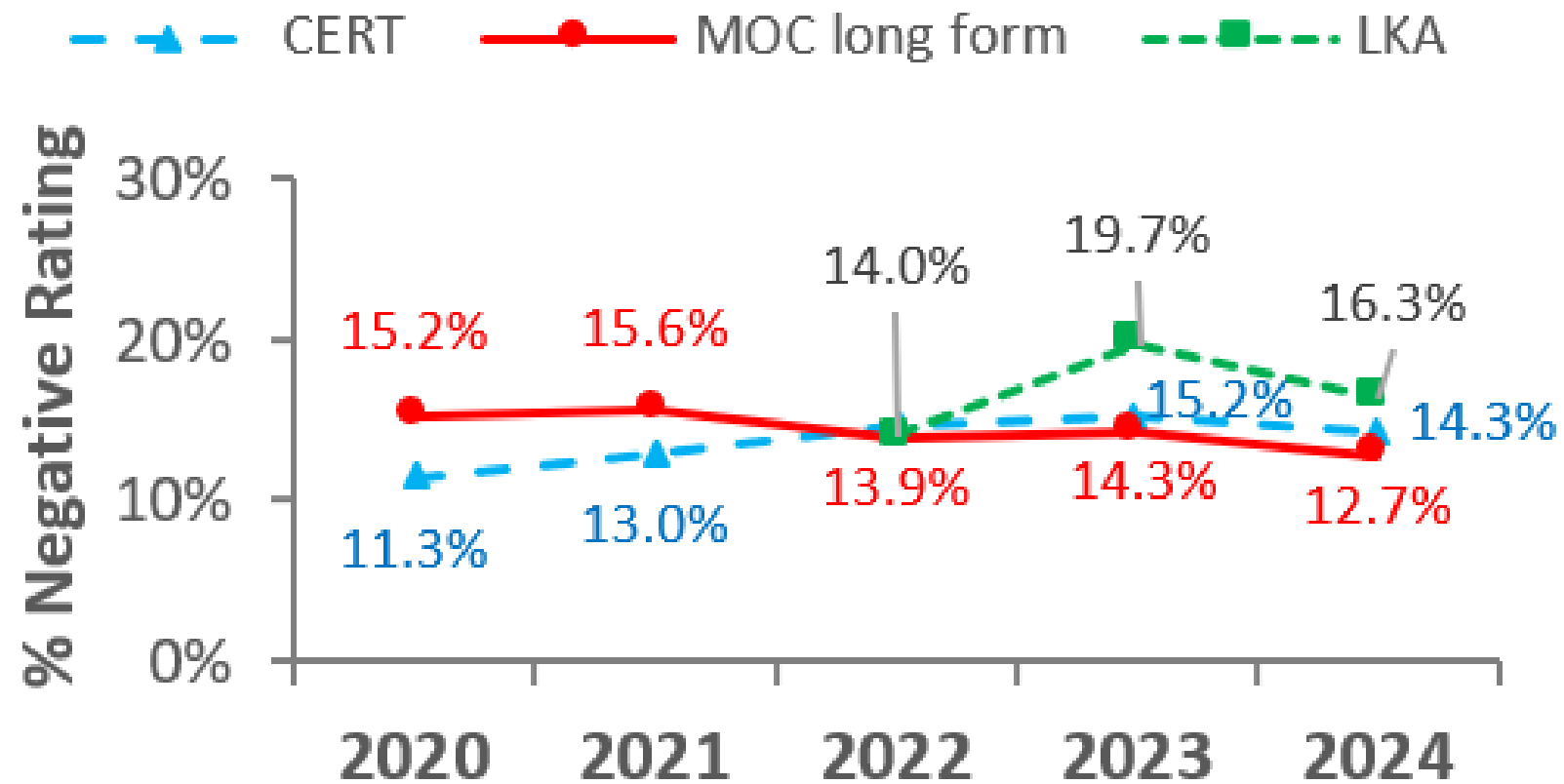
1. “Fair assessment” question is meant as gestalt: **questions, scope, difficulty**
2. Physicians can also rate assessment **format, and/or ABIM MOC itself** (“Is a timed, solo multiple-choice exam a fair way to assess medical knowledge?” “Is it fair that I have to do this at all?”)
3. LKA takers’ fairness ratings **don’t seem to be overly driven by their performance**
4. Physicians don’t necessarily bracket “clinical knowledge” from **other aspects** of assessments (technical, format, burden, learning; domains that are/not included) - these can all be seen as ***integrally connected and relevant***

Possible next steps

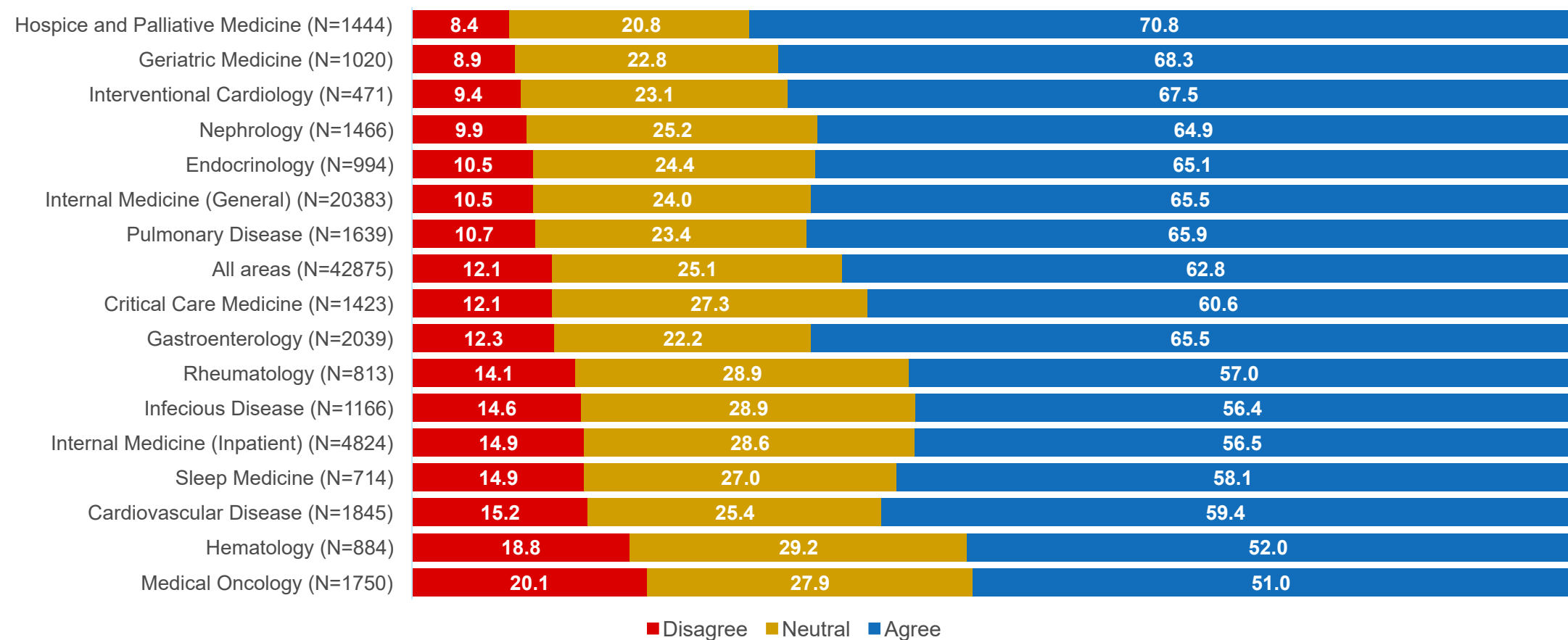
- New question on “What could ABIM do to make this a more fair assessment of clinical knowledge?” (*added to LKA 2025 end-of-year survey*)
- Publish findings (*in progress*)
- Further research – more disciplines; initial certification exam; survey with larger, more systematic sample; item writers and Approval Committee members
- “Fairness rubric” (what are other boards already doing?)
- Experiment with additional, focused questions on specific dimensions of “fairness”
 - e.g., “Should any board-certified physician in this field know this content?”
 - LKA surveys; full-day exam surveys; LKA post-item feedback; item writer training?

APPENDIX

EFFECTIVENESS: The examination was a fair assessment of clinical knowledge in this discipline



“I am satisfied with my overall LKA experience so far” survey responses*



*Based on the 2024 LKA End-of-Year Evaluation Survey, which had a 61% response rate. Numbers may not add up to exactly 100% due to rounding.

Interviews with staff leaders

“Fair assessment” question is intended to measure a **gestalt** of 3 domains related specifically to assessment content

- **Content of individual questions** – were there too many questions that felt irrelevant or esoteric?
- **Scope of content** – did the assessment cover the right topics to represent knowledge in the specialty?
- **Difficulty** – was it appropriately difficult?

Respondents have incomplete information on all of these at time of survey

- Long-form: haven’t received score information; are likely focused on most recent / memorable test items
- LKA: haven’t seen entire cycle’s set of questions; have received immediate feedback but may not have received / reviewed Progress Reports; may be more focused on recent quarter

Value of question may lie in its continuity over time – whatever it is measuring, it can be useful to compare specialties to each other, observe trends year-to-year

Suggested question – **“What could ABIM do to make this assessment more fair?”** – for interviews, LKA survey

	LKA 2023 disagree (300)	LKA 2023 neutr (200)	Long 2019-23 (330)	Total (830)
Scope of content	177	85	270	532
Questions (missing info, tricky, etc.)	133	94	103	270
Exam format (MCQs, security)	15	14	39	68
Timed aspect	30	17	1	48
MOC requirements	14	2	7	23
Too difficult	7	7	9	23
Pretests	12	8	n/a	20
Not useful for learning	11	0	n/a	11
Summative timing	0	1	6	7
Too burdensome	2	1	1	4
Tech issues	1	0	2	3
Too easy	2	1	0	3
Too early to assess	0	3	n/a	3
Progress report timing	0	1	n/a	1
?	8	5	6	19

Analysis of sample survey comments

“Biased against me” – respondents can perceive LKA to be biased against their own particular practice profile (whatever that is), e.g., 2 back-to-back **disagree** comments from IM Longform 2023 survey:

“Inpatient medicine was under-represented.” / “Seemed biased toward inpatient medicine.”

Comments from respondents with different fairness ratings can sound very similar:

- *“For the most part, yes [it is fair]. But... some questions didn't give us relevant information that we almost certainly would have had in real life...”* (IM LF 2022, **disagree**)
- *“For internists in primary care, the number of questions with a leader: ‘...the patient comes to the ER with...’ I really only counted one question on actual diabetes management.”* (IM LF 2021, **neutral**)
- *“There were a lot of questions on uncommon diagnoses, and on subspecialty management... Could have represented common diagnoses and management strategies better.”* (IM LF 2021, **agree**)

Hypothesis: “Similar criteria, different standards”

- Physicians may observe the same aspects of the assessments, but apply their individual standard when determining “fairness” (i.e., how many problematic items make something “unfair”?)

LKA Group	Invites Sent	Interviews Completed	Response Rate	Fairness Positive	% of pop.	Fairness Neutral	% of pop.	Fairness Negative	% of pop.
Internal Medicine	180	43	24%	18 (42%)	59%	12 (28%)	27%	13 (30%)	15%
Oncology	55	18	33%	5 (28%)	49%	6 (33%)	27%	7 (39%)	24%

Age (IM)		%	% of pop.
40-49	11	26%	35%
50-59	20	47%	38%
60-69	10	23%	21%
70-79	2	5%	3%

Age (ONCO)		%	% of pop.
40-49	3	17%	41%
50-59	8	44%	34%
60-69	7	39%	21%
70-79	0	0%	4%

Gender (IM)		%	% of pop.
Male	21	49%	49%
Female	22	51%	51%
Gender (ONCO)		%	% of pop.
Male	14	78%	60%
Female	4	22%	40%

Cert Combinations (ONCO)		%	% of pop.	
IM + ONCO		10	56%	38%
IM + ONCO + HEMA		8	44%	62%

Region (IM)		%	% of pop.
West	15	35%	25%
Midwest	4	9%	19%
South	13	30%	33%
Northeast	11	26%	23%

Region (ONCO)		%	% of pop.
West	8	44%	22%
Midwest	1	6%	18%
South	4	22%	35%
Northeast	5	28%	24%

- The response rate and number of interviewees are **very high** for projects of this type
- The interviewee samples reasonably resemble the overall populations
- “Resemble” ≠ “representative”... interview samples are not designed for generalization or estimating prevalence

Physician interviews – findings

Role of comprehensive knowledge assessment – at start of training vs independent practice

- Critics of LKA fairness often refer to initial cert exam as appropriate –
- A comprehensive knowledge exam to test one's basic fund of knowledge makes sense when just starting out, pre-specialization and pre-independent practice
- **Expectations for MOC assessment are different – physicians can question the value of what is “in” the assessment, given their awareness of what is “out”**
 - Practice specialization
 - Local context
 - Competencies other than knowledge – teamwork, communication
 - Patient outcomes

Physician interviews – portrait 1

Started Medical Oncology LKA in 2023; **disagreed** it is a fair assessment of clinical knowledge in the discipline

Male; age 43; highly specialized practice (treats only pancreatic cancer)

Treats LKA as *“a bit of an exercise”*; it covers content that will never be relevant to his subspecialized practice

Largely positive about LKA overall (scope of questions, difficulty)

Contrasts LKA with “pretty much every test I’ve ever taken”:

- **It’s not about choosing the right resources for intensive preparation beforehand**
- *“The bar is lower”* (because LKA is open book)

Notes that **pretest questions are a more “mentally present” aspect of LKA** compared to full-day exam

- Full-day: *“You are just trying to get through the day”*, not thinking about individual questions
- LKA: *“You might be doing just that one, or stop after and then think... ‘did that count?’”*

Physician interviews – portrait 2

Started Medical Oncology LKA in 2022; **neither agreed nor disagreed** it is a fair assessment

Male; age 62; retired 1 year ago from general hematology/oncology practice (now consults)

Familiarity / experience of participating in LKA (using the right info. resources, answering questions online, etc.) informs feelings about assessment & content

His general definition of a “fair assessment”: ***“It’s fair to push people a little outside of their comfort zone, as opposed to asking about what’s in a separate specialty people aren’t trained in”*** (e.g., transplant)... *“benign hematology, bleeding disorders, malignant hematology, that’s all fair game, everyone should have been exposed to that...”*

- *“We all have our strengths and weakness because of what we see routinely or not as often”*... for example, he struggles with hemoglobinopathies and never sees those patients, but it’s still fair to ask him questions on them: *“I’m supposed to be an expert in oncology.”*

Physician interviews – portraits 3 & 4

Started Internal Medicine LKA in 2024; **neither agreed nor disagreed** it is a fair assessment

Female; age 61; clinician / educator in NYC academic center

LKA format is connected to its fairness: “[LKA] fits the purpose, which is you’re trying to measure if an internist has a working foundation of knowledge. And what you’d do in practice with a real patient doesn’t always match [the textbook answer]”... but it’s fair to be tested on the “textbook answers” nonetheless.

Started IM LKA in 2023; **agreed** it is a fair assessment

Male; age 44; 100% telemedicine practice treating ADHD and menopausal patients

Welcomes broadness of topics and questions in areas that are unfamiliar areas or outside of his specialized practice – he learns from these.

- “Maybe, if anything, broaden the variety of the questions”... as opposed to narrowing them.
- Thinks full-day exam is inherently less fair than LKA: “After the first hour or two [of testing] you’re not working or thinking the way you do in practice.”

Physician interviews – implications

What are the dimensions of what makes an assessment “fair”? Does it...

- Represent **accurate** knowledge, **up-to-date** with current evidence?
- Focus on **substantive topics** that matter to patient care/outcomes, and common enough to be relevant (or if rare, important enough to test on)?
- Appropriately **represent the specialty** –
 - Nothing too subspecialized or out of field?
 - **Topics that everyone should know**, regardless of subspecialization?
 - **Balanced** across topics (e.g., not too heavy on some cancers vs others)?
- Feel **true to practice**?
 - (time limit, access to resources, solo thought process)
- Impose a **reasonable burden** on takers?
 - (frequency, flexibility, technical, transparency of standards)
- Help **build knowledge and improve practice**?
- **Evolve over time** to demonstrate ABIM is aware of MD concerns, striving to improve?

Physician interviews – implications

The fairest possible assessment:

- Asks about **topics that represent my practice / my specialty** as a whole
 - Things everyone in field should reasonably be expected to know to provide baseline care, regardless of practice subspecialization
- Captures what's important for **any physician in the field to know and be able to act on, on their own**
 - Knowledge physicians should have “at hand”, not outsourced to colleagues, EHR, UpToDate, etc.
- Minimal/no questions on topics **outside scope** of specialty
- Minimal/no questions that are **out of date**
- Minimal/no questions **written to be “tricky”** (purposefully deceive in a way real patients don't)
- **Educationally useful** (takers learn things they can apply to patient care)
- Imposes a **reasonable burden** on takers –
 - “The time and stress this requires is manageable, and worth the value I / my patients / society get from it”

Draft questions on dimensions of “fairness”

- *“Is the content accurate & up to date?”*
- *“Are the patient scenarios realistic?”*
- *“Should any board-certified physician in this field know this content?”*
- *“Are the questions on topics/scenarios that make a difference to patient care?”*
- *“Does the format (timed MCQs, answered alone) fairly assess a physician’s clinical knowledge?”*
- *“Are the requirements of participation (time, stress) reasonable for a practicing physician?”*

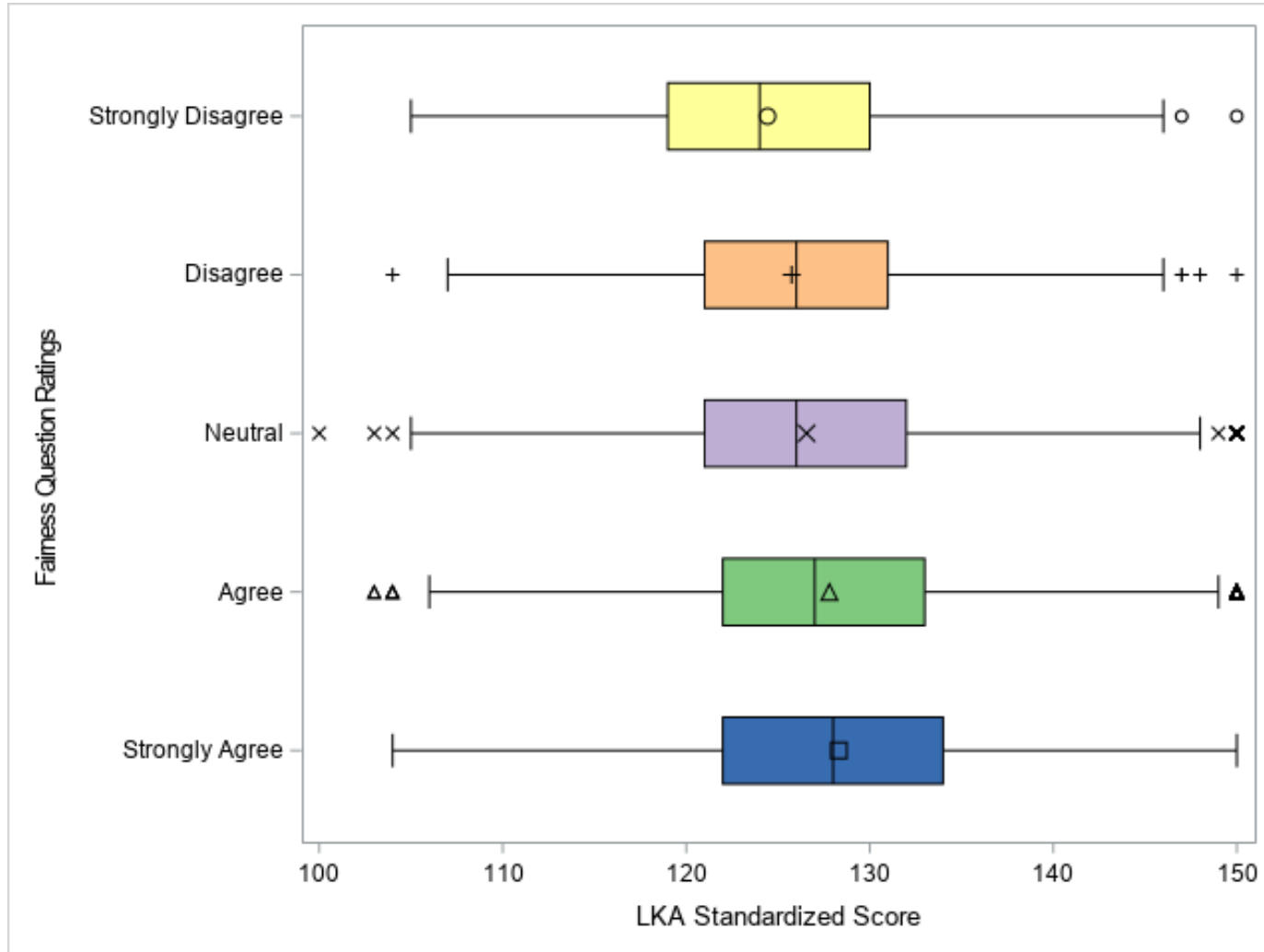


LKA Performance and Fairness Question Responses

LKA Participants Without Progress Reports – 2024 Cohort

IM 2024 Cohort

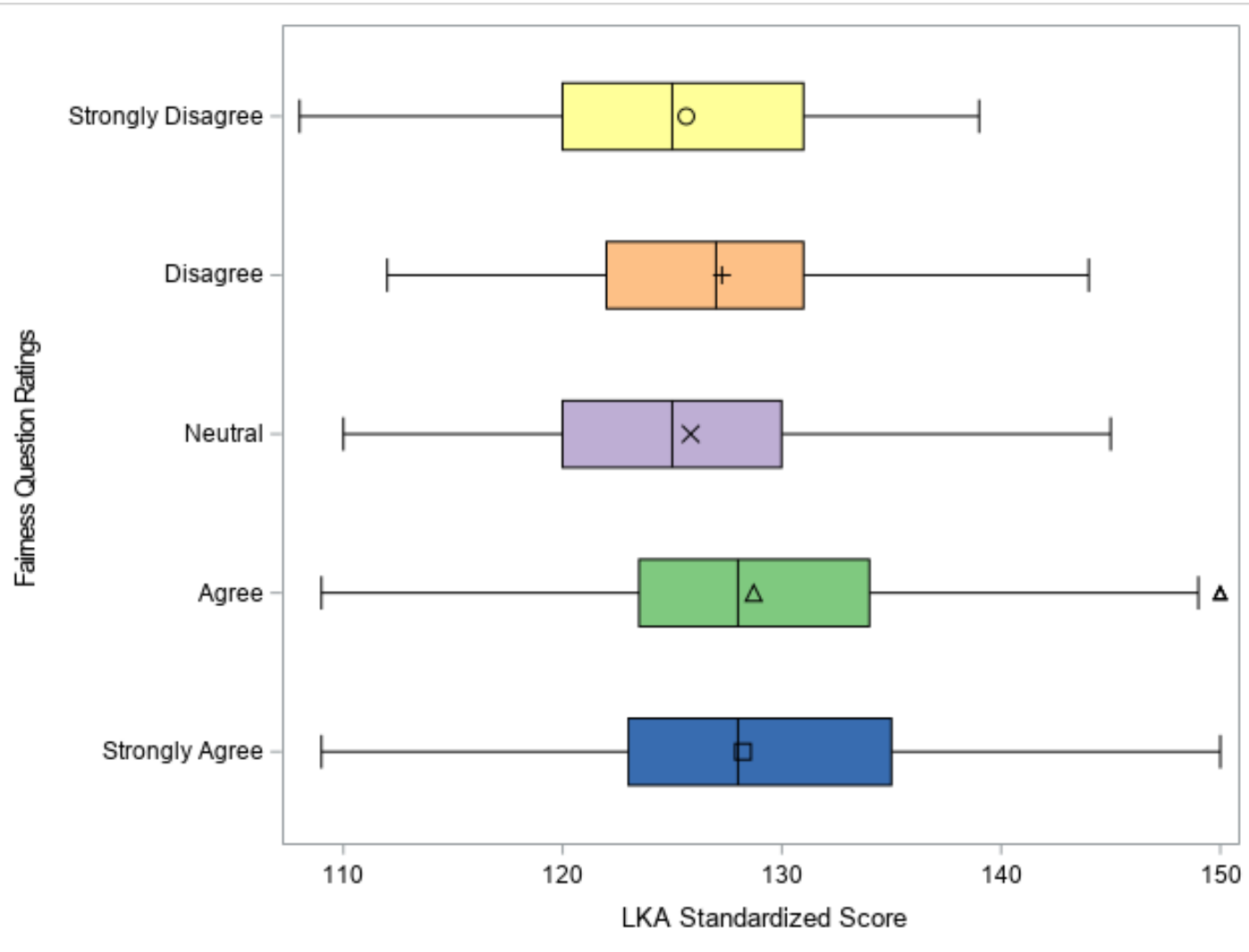
- Spearman Correlation: 0.1145 ($p < 0.05$)
- Weak positive correlation between LKA standardized scores and fairness question ratings



Fairness Question Rating (N)	Median LKA Standardized Score
Strongly Disagree (292)	124
Disagree (577)	126
Neutral (1,517)	126
Agree (2,647)	127
Strongly Agree (665)	128

ONCO 2024 Cohort

- Spearman Correlation: 0.1137 ($p < 0.05$)
- Weak positive correlation between LKA standardized scores and fairness question ratings



**Fairness Question
Rating (N)**

**Median LKA
Standardized Score**

Strongly Disagree (51)

125

Disagree (82)

127

Neutral (157)

125

Agree (208)

128

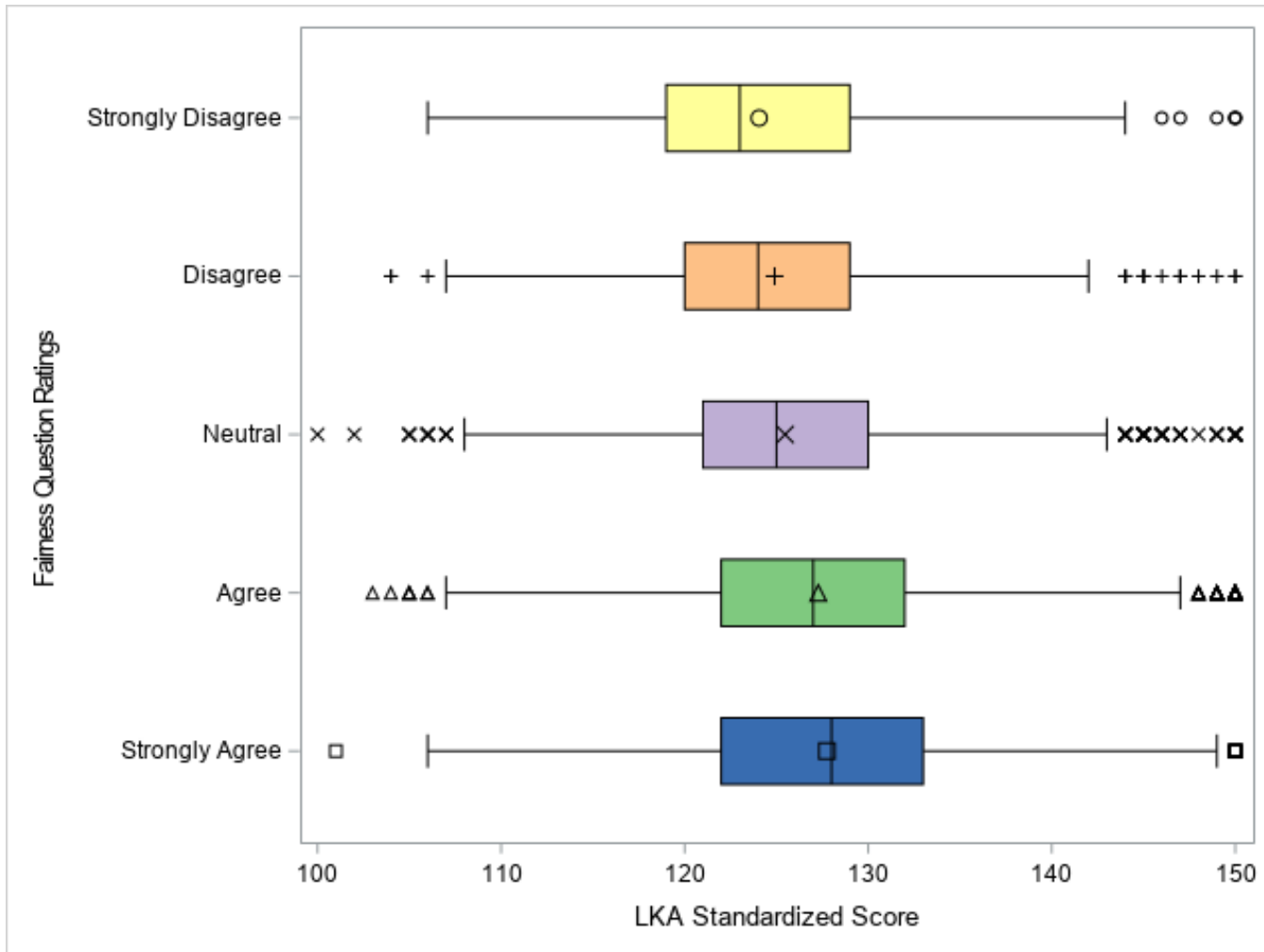
Strongly Agree (53)

128

LKA Participants With Progress Reports – 2022 and 2023 Cohorts

IM 2022 and 2023 Cohorts

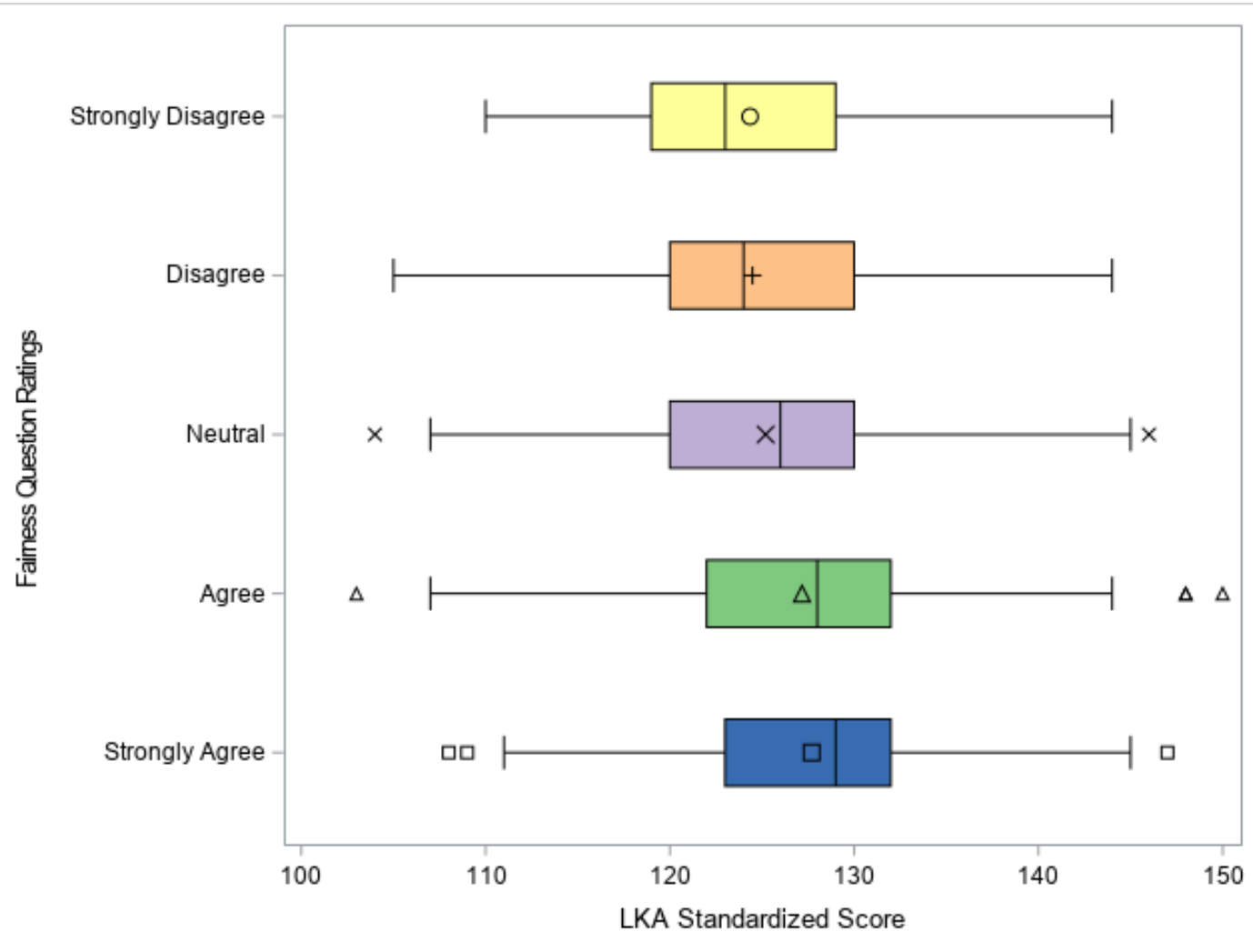
- Spearman Correlation: 0.1470 ($p < 0.05$)
- Weak positive correlation between LKA standardized scores and fairness question ratings



Fairness Question Rating (N)	Median LKA Standardized Score
Strongly Disagree (705)	123
Disagree (1,301)	124
Neutral (3,645)	125
Agree (6,575)	127
Strongly Agree (1,655)	128

ONCO 2022 and 2023 Cohorts

- Spearman Correlation: 0.1704 ($p < 0.05$)
- Weak positive correlation between LKA standardized scores and fairness question ratings



Fairness Question Rating (N)	Median LKA Standardized Score
Strongly Disagree (100)	123
Disagree (167)	124
Neutral (290)	126
Agree (431)	128
Strongly Agree (116)	129

Conclusions

- There is a weak positive relationship between LKA standardized scores and fairness ratings.
- The relationship between LKA standardized scores and fairness ratings does not differ meaningfully between LKA takers who received progress reports and those who did not.
- Factors other than participants' LKA performance may play a more substantial role in shaping their views on LKA fairness.